REAL BRAIN DATA TIME SERIES

Data description

Data are brain signals recorded via resting-state functional MRI (fMRI). Each of the two matrices contains 1200 time points for 119 brain regions.

↳ The two matrices correspond to two recording sessions of the same subject (ideally could be used to verify consistency of estimates).

Tasks

1. Calculate transfer entropy matrix of size 119×119 where element (i,j) represents the TE from the ith signal to the jth signal
2. Compare matrices obtained using different methods
3. Evaluate the CONSISTENCY of the results across the two sessions.

NOTE: If the computation is too heavy, you can restrict yourself to a SUBSET OF THE TIME SERIES
e.g. first 10 or even only first 2 regions.

Binning

For TE you need things like: $H(X_t | X_{t-1}, S_{t-lag})$
With binning we approximate $p(x_t, x_{t-1}, s_{t-lag})$ using an histogram!

Recall: TOO FEW BINS → many values fall in the same bin and you lose structure
(see METHODS_MI notes)

TOO MANY BINS → few samples per bin and many empty bins, TE becomes unstable

⚠ Curse of dimensionality now becomes relevant

For TE we're effectively building:

$H(X_t | X_{t-1}) \rightarrow$ 2D histogram
$H(X_t | X_{t-1}, S) \rightarrow$ 3D histogram

We have N samples, n_{bins}^d cells.
On average, per cell we have # samples $= N / n_{bins}^d$

For entropy estimation to be meaningful every cell should have at least $\sim\Theta(1)$ samples.

Imposing: $\frac{N}{n_{bins}^d} \approx 1 \Rightarrow n_{bins}^d \approx N \Rightarrow n_{bins} \approx N^{1/d}$

(at least this way you don't create more cells then data can fill)